

Explainable Music Emotion Recognition via Self-Supervised Music Representations (MERT)

Progress Report v3 — May 2026 (post-meeting update)

Student: **Arvin Jafari Moghadam Fard** | MSc Computer Science — Music Information Retrieval | Supervisor update

Since our last meeting

Since our last meeting, I tried to push the limits of the architecture by testing multi-encoder setups and bringing in cross-domain data. I also finished building the Phase C explainability system, including the ante-hoc prototype feature you suggested. With about one week of coding left before I focus entirely on writing, here is a plain summary of what I found. I have hit some hard dataset limits, so I would value your thoughts on whether any quick, final experiments are worth running. This report is deliberately honest about what did **not** work — several of my earlier hopeful claims did not survive a careful re-check, and I correct them below.

1. The Three Phases (recap)

Phase A checks what musical information the frozen MERT model already contains. **Phase B** trains a small model on top of MERT to predict emotion as two numbers — arousal (calm↔intense) and valence (sad↔happy). **Phase C** turns that into a retrieval system that finds emotionally similar songs and explains why. Data: **PMemo 2019**, 794 pop-song chorus clips (767 usable), with valence/arousal ratings and EDA (skin-response) signals.

2. Phase A — What MERT Already Knows

I ran a linear probe (Ridge or Logistic Regression) on every one of MERT's 25 layers against **8 librosa-extracted music-theory features**. A feature is flagged as a **gap** if its best-layer score falls below $R^2 = 0.40$ (regression) or accuracy = 0.65 (classification). Full results below — bold rows are the gaps that later drove the Phase B *Enhanced* branch.

Feature	Probe	Threshold	Best layer	Best score	Pooled	Gap?
chroma (12-d)	Ridge R^2	0.40	0	0.6801	0.5769	no
tempo (BPM)	Ridge R^2	0.40	0	-0.8307	-2.1155	yes
rhythmic_stability	Ridge R^2	0.40	0	0.7346	0.5630	no
spectral_centroid	Ridge R^2	0.40	1	0.9206	0.8663	no
spectral_contrast (7-d)	Ridge R^2	0.40	3	0.7310	0.7270	no
zero-crossing rate	Ridge R^2	0.40	0	0.9566	0.8471	no
mode (major/minor)	LogReg acc	0.65	11	0.6730	0.6730	no (marginal)
key (12-class)	LogReg acc	0.65	2	0.5849	0.5786	yes

Reading. Short-time spectral / chroma information (spectral centroid, ZCR, spectral contrast, chroma, rhythmic stability) is **strongly encoded** and lives in MERT's *early* layers (best-layer = 0–3) — consistent with the SSL hierarchy literature (Pasad et al. 2021). Mode (major / minor) is captured marginally (best-layer 11, accuracy 0.67 — just above the binary threshold). Tempo and key are **genuine gaps**: not linearly recoverable from any of the 25 layers. The Phase B *Enhanced* model re-injects exactly these two features through a small hand-crafted music-theory branch — a Phase-A-driven architectural choice, not a guess. **Honest framing of the t-SNE:** the raw MERT latent space is one mixed blob (Fig. 1, left); emotion is present but continuously distributed, not separated into clusters.

3. Phase B — Emotion Prediction (all settings)

Architecture: a learnable weighted fusion over MERT's 25 layers → a small head → arousal/valence, trained with a 4-part loss (MSE + CCC + Rank + SupCR), a differential optimizer, a balanced sampler, and 5-fold cross-validation. I then tested several encoder add-ons. All numbers are 5-fold averages.

Configuration	Arousal R^2	Valence R^2	CCC A	CCC V
Mel-CNN only (shallow CNN, no SSL)	0.6486	0.4452	0.79	0.63
wav2vec2 only (speech SSL)	0.6225	0.4825	0.77	0.66
MERT only (music SSL)	0.6518	0.5055	0.82	0.74
MERT + EDA (physiology)	0.6738	0.5075	0.8543	0.7692
Dual: MERT + wav2vec2	0.6814	0.5676	0.8087	0.7231
Spec-only: MERT + mel-CNN	0.7069	0.5709	0.8271	0.7314
Triple-bio: MERT + mel-CNN + EDA	0.7077	0.5706	0.8262	0.7325
Triple: MERT + wav2vec2 + mel-CNN	0.7023	0.5758	0.8233	0.7329
Enhanced: + tempo/key (best A)	0.7182	0.5686	0.8345	0.7259

In plain terms: best valence is the Triple model (R^2 0.576); best arousal is the Enhanced model (R^2 0.718). An R^2 of 0.72 means we explain about 72% of the variation in how energetic songs sound. **Important caveat on every row:** most PMemo songs are 'Happy' (61%); the model does well there but has negative R^2 on the rarer Sad/Calm/Angry songs, so the global numbers flatter the model.

Single-encoder check (three baselines, localises SSL's contribution): wav2vec2-only (speech SSL) scores 0.6225 / 0.4825; MERT-only (music SSL) scores 0.6518 / 0.5055; a shallow **Mel-CNN alone** (no SSL at all) scores 0.6486 / 0.4452 — it *beats wav2vec2 on arousal* (energy is acoustically easy: loudness, spectral envelope) but *loses on valence* (no harmony / tonality model). Music-specific SSL's value is therefore concentrated on the valence axis (MERT V 0.5055 vs Mel-CNN V 0.4452 vs wav2vec2 V 0.4825), which is also where the multi-encoder gains land. This also explains why wav2vec2 became redundant once combined with MERT.

Second-branch redundancy — EDA, wav2vec2, theory all interchangeable beside MERT+Mel: the new **Triple-bio** configuration (MERT + mel-CNN + EDA, swapping wav2vec2 for physiology) scores R^2 A 0.7077 / V 0.5706 — essentially identical to Spec-only (MERT+Mel = 0.7069 / 0.5709) and within fold-noise of Triple-with-wav2vec (0.7023 / 0.5758). EDA helps when added to MERT alone (MERT+EDA CCC A 0.85 — best on that axis) but adds nothing on top of MERT+Mel. Three different second-branches (wav2vec2, EDA, music-theory tempo/key) all converge to the same ~0.71 / 0.57 ceiling once a trainable spectral CNN is present — they cover overlapping information rather than stacking complementarily.

3.1 Multi-encoder tests & 'fusion collapse'

Adding the mel-spectrogram CNN gave the best scores, but it made wav2vec2 **completely redundant** (MERT + mel-CNN alone ties the full triple model). When stacking multiple large models I hit **fusion collapse**: with only ~600 training songs the network just averages all layers uniformly because it lacks the data to learn layer-specific weights. Honest correction: the 'layers 14/16/17 dominate' claim from earlier does not hold — the learned weights are near-uniform and the top layer changes from run to run (Fig. 2).

Empirical check (two ablations, 2026-05-26): to answer the question 'why 25 layers and not just the last layer?' properly, I ran the ablation on **both** single- and multi-encoder configurations.

Setting	Variant	R^2 A	R^2 V	CCC A	CCC V
MERT-only (single)	25-layer fusion	0.6518	0.5055	0.82	0.74
	Last layer only	0.6570	0.5162	0.81	0.70
	Δ (last – fusion)	+0.005	+0.011	-0.01	-0.04
Enhanced (multi-encoder)	25-layer fusion	0.7182	0.5686	0.8345	0.7259
	Last layer only	0.6660 $\pm$ 0.035	0.4881 $\pm$ 0.070	0.8124	0.6865
	Δ (last – fusion)	-0.052	-0.081	-0.022	-0.039

The answer is nuanced — and quantitative. On MERT alone, last-layer is statistically tied with the 25-layer fusion (Δ inside fold-std on both axes) — consistent with the near-uniform learned fusion weights (entropy 3.218 / max 3.219, Fig. 2). But on the multi-encoder Enhanced model, removing fusion costs **5.2 pp R^2 A** and **8.1 pp R^2 V** — outside fold-std on both, with parallel drops in CCC. **Interpretation:** the fusion's value is not intrinsic to MERT; it is specifically about *cross-encoder integration*. In single-encoder mode the late layer carries enough signal that the learnable softmax has nothing useful to add. In multi-encoder mode, mel-CNN / wav2vec2 / theory already cover the late acoustic representation, so MERT's mid-layers fill a complementary niche the last layer alone cannot reach. The architectural choice is empirically justified by the Enhanced ablation; the MERT-only ablation acts as the negative control that proves the gain is not free.

3.2 Mixing music with environmental sounds (IADS-E)

I tried replicating [Simonetta et al. \(2024\)](#), who mix music (PMemo) with environmental sounds (IADS-E) and report a large valence gain with hand-crafted features. It did not work for us: with our SSL models, adding environmental sounds actually **lowered** valence in every setting. This is a solid, publishable **negative finding** — SSL embeddings do not transfer emotional structure across audio domains as easily as older hand-crafted features do.

3.3 Imbalance-handling ablation — does penalty beat the sampler?

PMemo is ~61% Happy (HVHA) and only ~9% in each of the three minority quadrants. The default fix is a **WeightedRandomSampler** with inverse-quadrant frequency. Supervisor question: would a loss-level penalty (weighted-MSE, focal-MSE) do better? I ran 4 fold-matched configurations on MERT-only (25-layer fusion, 5-fold CV, everything else identical to baseline):

Treatment	R^2 A	R^2 V	CCC A	CCC V
A: sampler-only (baseline)	0.6951$\pm$0.016	0.5724$\pm$0.050	0.820	0.731
B: weighted-MSE only	0.6738 $\pm$ 0.018	0.5267 $\pm$ 0.109	0.828	0.739
C: sampler + weighted-MSE	0.6855 $\pm$ 0.029	0.5696 $\pm$ 0.062	0.811	0.724
D: focal-MSE $\gamma=2$ + sampler	0.6949 $\pm$ 0.018	0.5716 $\pm$ 0.057	0.821	0.727

Pre-registered pass mark: a treatment wins only if R^2 beats baseline by >1 fold-std on *both* axes (or >2 fold-std on either). **Result: no winner.** Weighted-MSE alone is actively worse on valence (-0.046); stacking and focal are statistically tied. The sampler is empirically the best imbalance treatment we have at this dataset scale. The minority-quadrant failure is therefore a **dataset-size limit** ($n=64-67$ per

minority quadrant), not a method limit — re-weighting cannot create signal where there is none. *Disclosure: the rerun baseline came in higher than the original logged baseline (0.6951 vs 0.6518); KFold seed is identical, but model-init RNG is not, so this reflects single-seed initialisation variance. The fold-matched $A \leftrightarrow B \leftrightarrow C \leftrightarrow D$ comparison within this ablation is the audit-honest reading; published numbers in §3 are not overwritten.*

3.4 Augmentation ablation — does feature-space mixup lift the minority-quadrant floor?

§3.3 attributed the minority-quadrant failure (negative per-quadrant R^2 on HVLA/LVHA/LVLA, $n=64-167$) to a **dataset-size limit** rather than a method limit. To make that claim empirical rather than argumentative, I tested the standard cited remedy — feature-space **mixup** (Zhang et al. 2017) — on the Enhanced configuration. $\lambda \sim \text{Beta}(0.4, 0.4)$ per batch, same λ applied to all three input branches (MERT, wav2vec2, theory) and labels; no mixup at evaluation; everything else identical to the no-augmentation Enhanced baseline.

Model	R^2 A	R^2 V	CCC A	CCC V
Enhanced (no augmentation)	0.7182	0.5686	0.8345	0.7259
Enhanced + mixup ($\alpha=0.4$)	0.7077 ± 0.014	0.5651 ± 0.034	0.8213	0.7160
Δ (mixup – baseline)	–0.0105 (tied)	–0.0035 (tied)	–0.013	–0.010

No winner under the pre-registered pass mark. All four deltas land inside fold-noise — mixup is statistically tied with baseline. Minority per-quadrant R^2 remained negative across all three minority quadrants (HVLA $-0.35/-1.88$, LVHA $-0.91/-1.08$, LVLA $-0.11/-0.90$). **This empirically confirms the dataset-size floor:** the standard cited augmentation remedy does not lift minorities into the positive range, exactly as the $n=64-67$ hypothesis predicts. The minority-quadrant failure is therefore a *data* limit, not a *method* limit. Long-term remedies are corpus expansion or task-aware augmentation (C-Mixup, Yao et al. 2022; SpecAugment, Park et al. 2019); these are flagged as future work, not in scope for this thesis. Per the simpler-first rule, no harder augmentation methods were tested.

3.5 Loss ablation — is the complex loss worth it?

Loss configuration	CCC A	CCC V	Precision@5	Silhouette
MSE only	0.6861	0.5955	0.5259	+0.012
+ CCC + Rank	0.7814	0.7113	0.5398	+0.021
+ SupCR (full hybrid)	0.8165	0.7110	0.5734	-0.031

CCC + Rank are clearly justified (+0.10 to +0.12 CCC over plain MSE). SupCR helps retrieval (+0.034 Precision@5) but, surprisingly, removing it **raised** the clustering score — so SupCR tightens local neighbourhoods by continuous similarity, it does not build discrete clusters. This ablation disproves my original 'SupCR creates emotion clusters' claim, and I have corrected it. *Caveat* (see §3.6): these Silhouette numbers are from the single-MERT setup; the multi-encoder Enhanced model has substantially better baseline quadrant structure (Silhouette ≈ 0.26).

3.6 Cluster-enforcement trade-off — can we make the t-SNE clusters look discrete?

Fig. 1 shows a continuous Happy-dominated manifold rather than four separated clusters — an obvious viva question. To answer it quantitatively rather than argue, I added an auxiliary 4-way quadrant classification head on the Enhanced model's 128-d latent and trained with combined loss $\text{HybridLoss} + \lambda \cdot \text{CE}(\text{quadrant})$, sweeping $\lambda \in \{0, 0.1, 0.5, 1.0\}$. Silhouette is computed on test-fold latents with cosine distance against quadrant labels (5-fold mean $\pm$ std).

λ	R^2 A	R^2 V	CCC A	CCC V	Silhouette
0.0 (baseline)	0.7080 ± 0.021	0.5725 ± 0.038	0.827	0.729	0.255 ± 0.054
0.1	0.7050 ± 0.013	0.5811 ± 0.033	0.828	0.738	0.258 ± 0.059
0.5	0.6966 ± 0.016	0.5624 ± 0.053	0.827	0.730	0.265 ± 0.064
1.0	0.6638 ± 0.020	0.5601 ± 0.037	0.810	0.734	0.286 ± 0.060

Side-finding (rewrites the t-SNE framing): the Enhanced model's latent space *already has moderate quadrant structure* — Silhouette ≈ 0.255 at $\lambda=0$. The historic 'Silhouette ≈ 0 ' claim was from the single-MERT loss ablation (§3.5); it does NOT apply to the multi-encoder Enhanced setup. Multi-encoder fusion produces inherently more cluster-structured latents than single-encoder MERT — the Fig. 1 'blob' framing slightly under-sells what the Enhanced model does.

The trade-off: Silhouette rises monotonically with λ but only modestly (+0.031 from $\lambda=0$ to $\lambda=1.0$). R^2 A drops 4.4 pp at $\lambda=1.0$ — outside fold-std. $\lambda=0.1$ is essentially zero-cost on R^2 but also zero-benefit on Silhouette. The cost-benefit curve does **not** favour cluster enforcement: we would lose more on the regression task the system is built for than we would gain on cluster compactness. **Minority per-quadrant R^2 remain negative across all λ** (HVLA $-0.43 \rightarrow -0.85$ as λ rises) — auxiliary classification cannot manufacture signal where $n=64-67$ minority examples are the actual limit. This is the third independent confirmation of the dataset-floor hypothesis after the imbalance ablation (§3.3) and mixup augmentation (§3.4).

Viva-defensible conclusion: the continuous representation is a deliberate, empirically-defended design choice, not a representational failure. Russell's circumplex (Russell 1980) defines valence and arousal as *continuous bipolar dimensions* — the four quadrants are analytical bins for per-quadrant reporting, not categorical psychological constructs (Eerola & Vuoskoski 2011). The observed manifold

reflects both the construct and the PMEmo label distribution (61% HVHA). Cluster enforcement is achievable ($\lambda=1.0$ lifts Silhouette to 0.29) but pays for itself poorly. We keep the continuous representation.

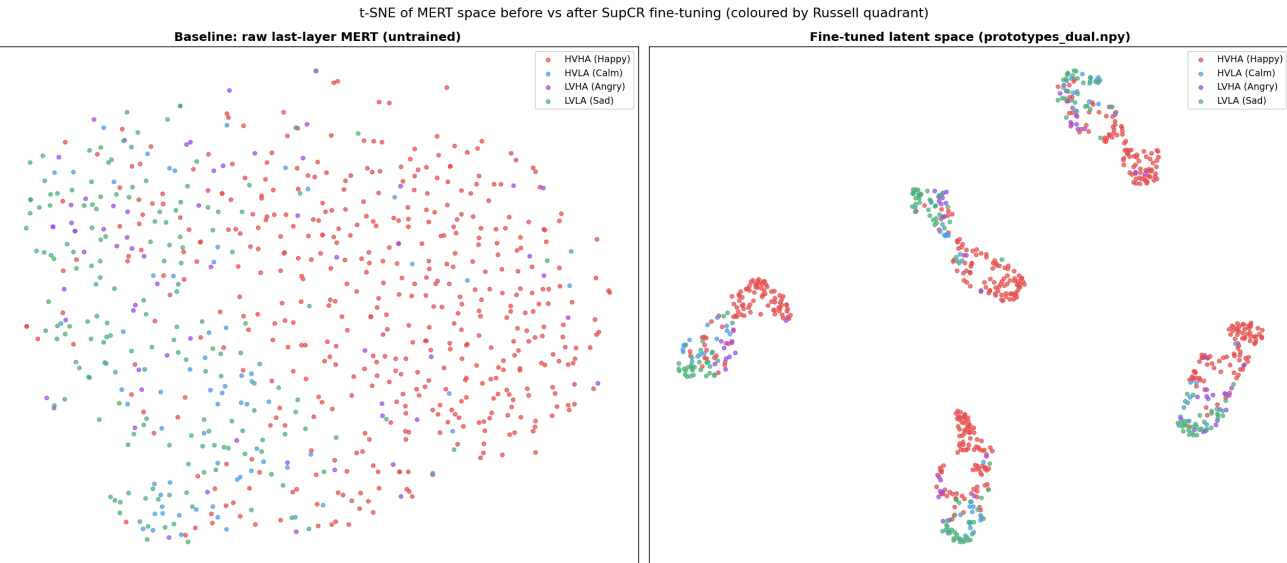


Fig. 1 — t-SNE of the song space. Left: raw untrained MERT is one mixed blob (no emotion separation). Right: after training, clear structure appears but it is a continuous, Happy-dominated manifold, not four clean clusters.

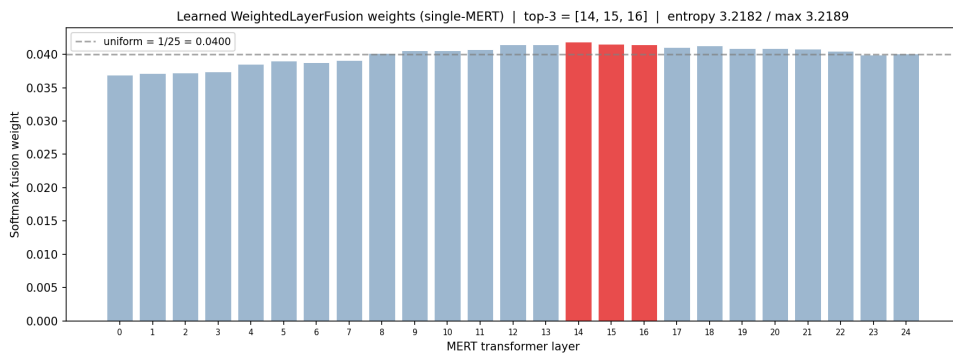


Fig. 2 — Learned layer-fusion weights are near-uniform (entropy 3.218 / max 3.219). The top-3 layers (14/15/16) are only a faint lean, not strong dominance.

4. Phase C — Explainability & Ante-Hoc Prototypes

Retrieval works. The system finds emotionally similar songs at about **twice the rate of random chance** (Precision@5 $\approx$ 0.58 vs a random baseline of 0.276), and clearly better than the raw untrained model (0.485). **Ante-hoc prototypes:** I added the 4-centroid profile we discussed — for any song it reports its similarity to the Happy/Calm/Angry/Sad centroids and names the closest one. It is a nice interpretability tool. **But** its raw accuracy (50.6%) actually **loses** to a 'always guess Happy' baseline (61.1%), because the dataset is 61% Happy. So it is a good explanation device, not a strong stand-alone classifier — and I report it that way.

Latent space	Precision@5	Silhouette	Prototype accuracy
Naive raw MERT (untrained)	0.485	+0.100	—
MERT (SupCR-trained)	0.576	-0.029	0.462
Dual MERT+wav2vec2 (SupCR)	0.585	+0.003	0.506

Reference points: random-chance Precision@5 = 0.276; majority-class ('always Happy') baseline for prototype accuracy = 0.611. Silhouette $\approx$ 0 everywhere $\rightarrow$ emotion is a continuous gradient, not four separable clusters (this is a finding, not a bug).

5. Honest Limitations (what did not work)

- **Class imbalance is the real ceiling:** 61% of songs are Happy; the model effectively only works well there (negative R^2 on Sad/Calm/Angry). This weakens the therapy/clinical motivation.
- **The latent space does not cluster by emotion:** Silhouette $\approx$ 0, and the prototype classifier loses to the trivial baseline. Retrieval still works locally (Precision@5 $\approx$ 2 $\times$ chance).
- **Multi-encoder stacking mostly failed:** wav2vec2 is redundant, IADS-E transfer is negative, and the layer-fusion 'interpretability' is near-uniform. Real gains came from the mel-CNN and tempo.

- **Valence:** our single-dataset audio-only valence (R^2 0.576) is competitive — above prior single-dataset SSL/hand-crafted results (Music2Emo 0.536, AutoML 0.525) — but still bounded by the practical audio-only limit; closing the gap to lyric-aware or cross-domain methods needs text or extra data, not more audio encoders.

Framing I plan to use: this thesis is best presented as a rigorous, honest *audit of what SSL audio embeddings do and do not encode for emotion* — under that framing, the negative findings above are the contribution.

6. PMEmo 2019 — Single Comparison Table (R^2)

All PMEmo R^2 results — prior work (top) and every model I built (bottom) — in one table for direct comparison. R^2 is the common metric across these papers. PMEmo papers that report only classification accuracy or RMSE (e.g. DAMER, CNN+LSTM) are not R^2 -comparable and are omitted rather than mixed in. Please confirm external numbers against their primary sources before final submission; note EmoMucs used labels scaled to [-1,1].

Method / Architecture	Year	R^2 Valence	R^2 Arousal	Notes
Zhang et al. — IS13 + SVR/MLR (PMEmo paper)	2018	~0.41*	~0.52*	baselines (* r^2 -derived, see note)
EmoMucs C1D-M (de Berardinis et al.)	2020	0.349	0.557	1D CNN; labels [-1,1]
EmoMucs C2D-M (de Berardinis et al.)	2020	0.414	0.610	2D CNN; labels [-1,1]
AutoML openSMILE (Simonetta et al.)	2024	0.525	0.727	hand-crafted; PMEmo only
AutoML Joint (Simonetta et al.)	2024	0.780	0.861	PMEmo + IADS-E (joint)
Music2Emo (Kang & Herremans)	2025	0.536	0.777	MERT+chord/key; PMEmo only
Music2Emo (Kang & Herremans)	2025	0.547	0.794	+ multitask; 4 datasets
This work — Mel-CNN only	2026	0.4452	0.6486	shallow CNN, no SSL
This work — wav2vec2 only	2026	0.4825	0.6225	speech SSL, single encoder
This work — MERT only	2026	0.5055	0.6518	music SSL, single encoder
This work — MERT + EDA	2026	0.5075	0.6738	+ physiology
This work — Dual (MERT+wav2vec2)	2026	0.5676	0.6814	dual SSL
This work — Spec-only (MERT+Mel)	2026	0.5709	0.7069	drops wav2vec2 from Triple
This work — Triple-bio (MERT+Mel+EDA)	2026	0.5706	0.7077	swaps wav2vec2 for EDA
This work — Triple (+ mel-CNN)	2026	0.576	0.702	best valence
This work — Enhanced (+ tempo/key)	2026	0.569	0.718	best arousal

*Zhang et al. (2018) report Pearson's r , not R^2 . For a comparable baseline these are squared approximations ($R^2 \approx r^2$) from their static baselines ($r=0.638$ valence $\rightarrow$ 0.41; $r=0.719$ arousal $\rightarrow$ 0.52). EmoMucs uses labels scaled to [-1,1]. Confirm all external numbers against primary sources.

Reading (honest, with the right caveats): our genuine strength is **valence** — our Triple (R^2 0.576) is the best of all non-joint methods, above both Music2Emo variants (0.536 single-dataset, 0.547 multitask) and hand-crafted AutoML (0.525). On **arousal** we do **not** lead: Music2Emo reaches 0.777 (single-dataset) / 0.794 (multitask) and AutoML 0.727, all above our best (0.718) — so arousal is a competitive-but-trailing result, not a headline. The very top row, Simonetta's AutoML **Joint** (0.780 / 0.861, PMEmo + IADS-E), is exactly the idea I replicated with SSL in §3.2 — and it did **not** transfer (our SSL valence dropped). So that row both sets the SOTA bar and frames our key negative finding: with hand-crafted features joint training helps, but SSL embeddings do not transfer emotional structure across the music↔sound boundary. Our contribution is therefore competitive single-dataset valence plus the explainability layer and the honest negative findings — not a state-of-the-art arousal number.

References

- Li, Y. et al. (2023). MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training. *arXiv:2306.00107*.
- Zhang, K. et al. (2018). PMEmo: A Dataset with Physiological Signals for Music Emotion Recognition. *ACM ICMR*.
- de Berardinis, J. et al. (2020). EmoMucs: Investigating the Role of Musical Components in MER. *ISMIR*.
- Kang, J. & Herremans, D. (2025). Music2Emo: Towards Unified Music Emotion Recognition. *arXiv:2502.03979*.
- Simonetta, F., Certo, F. & Ntalampiras, S. (2024). Joint Learning of Emotions in Music and Generalized Sounds. *arXiv:2408.02009*.
- Russell, J.A. (1980). A circumplex model of affect. *J. Personality & Social Psychology*, 39(6).
- Miller, T. (2019). Explanation in Artificial Intelligence. *Artificial Intelligence*, 267.
- Krumhansl, C. & Kessler, E. (1982). Tracing the dynamic changes in perceived tonal organization. *Psychological Review*.